Limit

✓ $a^2-b^2 = (a+b)(a-b)$

$\sqrt{(-2)^2} = |-2| = 2$

Definisi : Suatu pendekatan

1) $\lim_{x\to -3} \dfrac{x^3+27}{x+3} = \lim_{x\to -3} \dfrac{(x+3)(x^2-3x+9)}{(x+3)}$ ← elim bagian yg bikin error

$= \lim_{x\to -3} (x^2-3x+9)$

$= (-3)^2 - 3\cdot(-3) + 9$

$= 9 + 9 + 9 = 27$ //

$x^3+27 = (x+3)(x^2-3x+9)$

$x+3 \;\overline{)\; x^2-3x+9}$

x^3+27

x^3+3x^2 −

$-3x^2+27$

$-3x^2-9x$ −

$9x+27$

$9x+27$ −

0

	1	0	0	27
−3		−3	9	−27
	1	−3	9	0

↓

x^2-3x+9

2) $\lim_{x\to\infty} \left(x-a-\sqrt{x^2-2ax+b}\right) = \lim_{x\to\infty} \dfrac{(x-a)-\left(\sqrt{x^2-2ax+b}\right)}{1} \cdot \dfrac{(x-a)+\left(\sqrt{x^2-2ax+b}\right)}{(x-a)+\sqrt{x^2-2ax+b}}$

Limit infinite dikali. $\frac{\frac{1}{x^n}}{\frac{1}{x^n}}$

① coba substitusi dulu

② klo hasilnya error / invalid, kali sekawan

$= \lim_{x\to\infty} \dfrac{(x-a)^2 - |x^2-2ax+b|}{x-a+\sqrt{x^2-2ax+b}} = \lim_{x\to\infty} \dfrac{x^2-2ax+a^2 - x^2 - |-2ax+3|}{x-a+\sqrt{x^2-2ax+b}}$

2) $= \lim_{x\to\infty} \dfrac{-2ax + a^2 - |-2ax+a|}{x - a + \sqrt{x^2 - 2ax + a^2}} \cdot \dfrac{\frac{1}{x}}{\frac{1}{x}}$

$= \dfrac{-2a - |-2a|}{1+\sqrt{1}} = -2a - 2a = -4a$ //

+ sudut berelasi
+ sudut ganda

3) Trigon

$\lim_{x\to 0} \dfrac{\sin x}{x} = \lim_{x\to 0} \dfrac{\tan x}{x} = 1$

$\lim_{x\to 0} \dfrac{\sin ax}{x} = \lim_{x\to 0} \dfrac{\sin ax}{x} \cdot \dfrac{a}{a}$ (a)

$= \lim_{x\to 0} \dfrac{\sin ax}{ax} \cdot a = 1 \cdot a = a$ //

$\cot x = \dfrac{1}{\tan x} = \dfrac{\cos x}{\sin x}$

$\sec x = \dfrac{1}{\cos x}$

$\csc x = \dfrac{1}{\sin x}$

$\sin^2 x + \cos^2 x = 1$

$\tan^2 x + 1 = \sec^2 x$

$1 + \cot^2 x = \csc^2 x$

$\sin(x+y) = \sin x \cos y + \cos x \sin y$

$\sin(x-y) = \sin x \cos y - \cos x \sin y$

$\cos(x+y) = \cos x \cos y - \sin x \sin y$

$\cos(x-y) = \cos x \cos y + \sin x \sin y$

$\tan(x+y) = \dfrac{\tan x + \tan y}{1 - \tan x \tan y}$

$\tan(x-y) = \dfrac{\tan x + \tan y}{1 + \tan x \tan y}$

$$\lim_{x \to 0} \frac{\sin 4x}{x} = \lim_{x \to 0} \frac{\sin 4x}{(x \cdot 2)\frac{1}{2}}$$

$$= \lim_{x \to 0} \frac{\sin 2x}{2x} \cdot 2$$

$$= \lim_{x \to 0} 1 \cdot 2$$

$$= 2$$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

4) L'Hôpital

$\frac{0}{0}$ ou $\frac{\infty}{\infty}$ → L'Hôpital

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}$$

1. Find the value of the limits below!

a. $\lim_{x \to 2} 3x + 1$

b. $\lim_{x \to 0} (x^2 + 4x)$

c. $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

d. $\lim_{x \to 4} \sqrt{x}$

e. $\lim_{x \to 5} 7$

a) $\lim_{x \to 2} (3x + 1) = 3 \cdot 2 + 1 = 7$ //

b) $\lim_{x \to 0} (x^2 + 4x) = 0 + 4 \cdot 0 = 0$ //

c) $\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = \lim_{x \to 3} \frac{(x-3)(x+3)}{x-3} = \lim_{x \to 3} (x + 3) = 3 + 3 = 6$ //

d) $\lim_{x \to 4} \sqrt{x} = \sqrt{4} = 2$ //

e) $\lim_{x \to 5} 7 = 7$ //

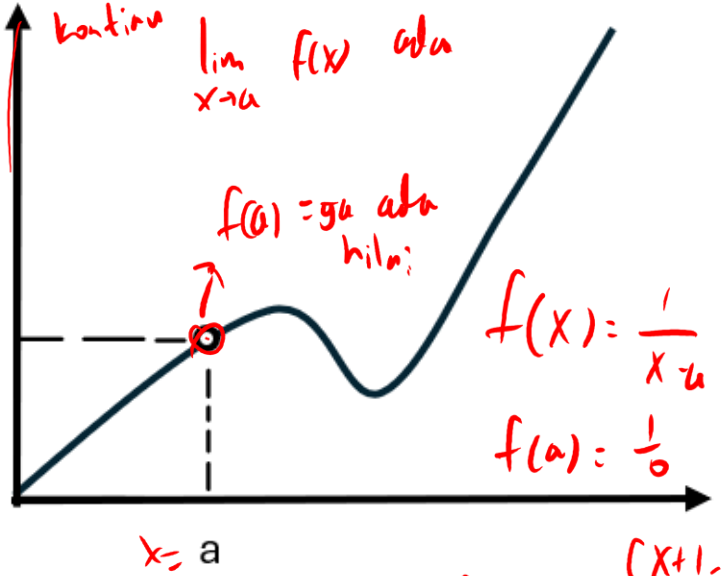
tidak kontinu

$\lim_{x \to a} f(x)$ ada

$f(a)$ = tak ada nilai

$f(x) = \frac{1}{x-a}$

$f(a) = \frac{1}{0}$

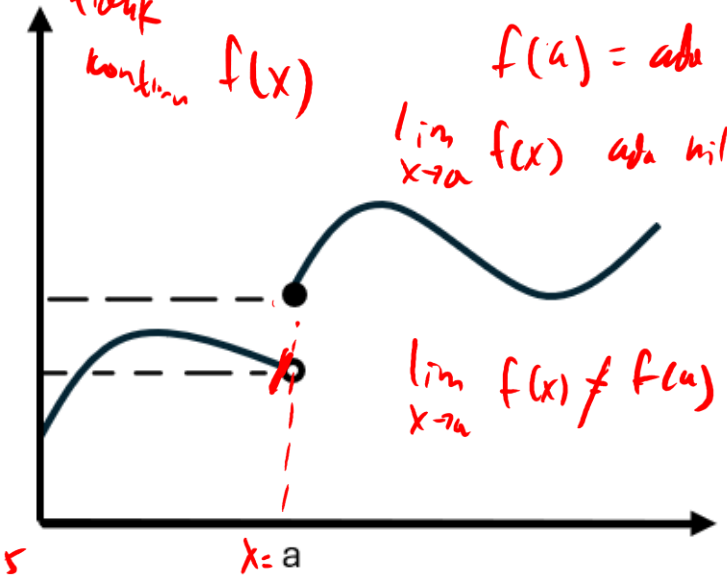


tidak kontinu $f(x)$

$f(a)$ = ada nilai

$\lim_{x \to a} f(x)$ ada nilai

$\lim_{x \to a} f(x) \neq f(a)$



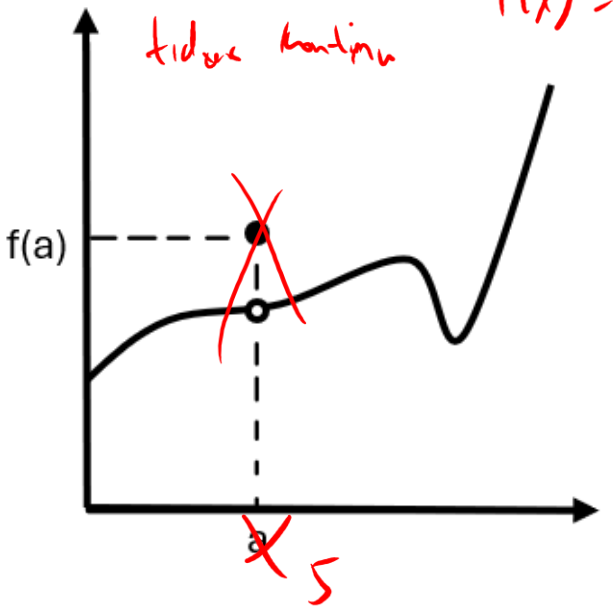
$$f(x) \begin{cases} x+5 & x \geq a \\ x-1 & x < a \end{cases}$$

$x - 1 \to a - 1$

$x + 5 \to a + 5$

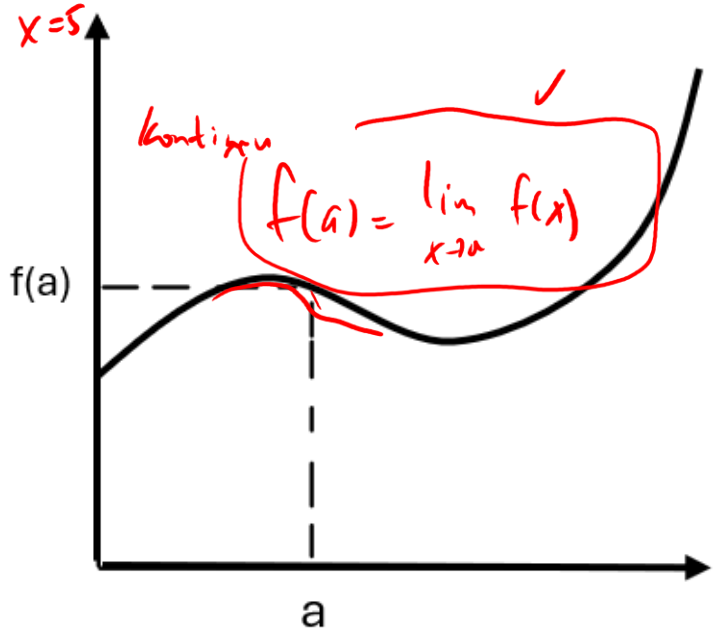
$$f(x) = \begin{cases} x+1, & x \neq 5 \\ x^2, & x = 5 \end{cases}$$

tidak kontinu



kontinu ✓

$f(a) = \lim_{x \to a} f(x)$



6. Evaluate using L'Hopital Rule :

a. $\lim_{x \to 1} \frac{x^2 + 16x - 17}{x - 1}$

$\frac{d}{dx} x^{(n)} = n \cdot x^{n-1}$

$\frac{1^2 + 16 \cdot 1 - 17}{1 - 1} = \frac{0}{0}$ → bisa pakai l'Hôpital

$\frac{d}{dt} S(t) = v(t)$ $\quad f(x) = s$

$\lim_{x \to 1} \frac{\frac{d}{dx}(x^2 + 16x - 17)}{\frac{d}{dx}(x - 1)} = \lim_{x \to 1} \frac{2x + 16}{1}$

$= 2 \cdot 1 + 16 = 18$

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h - 0}$ ($y_2 - y_1$; $h \downarrow x_2$, $0 \downarrow x_1$)

$\frac{\Delta y}{\Delta x} = m$ = gradien

(1, 18)

$\frac{d}{dx}(-17 \cdot x^{(0)}) = 0 \ldots = 0$

C. $\lim_{x \to 0} x \ln(2x)$

$\log a - \log b = \log \frac{a}{b}$

ln → natural logarithm → euler's number

L'Hôpital

$f(x) = x \ln(2x)$

$f'(x) = 1 \cdot \ln 2x + x \cdot \left(\frac{d}{dx} \ln 2x \right)$

$g(x) = \ln 2x$

$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$

$= \lim_{h \to 0} \frac{\ln(2(x+h)) - \ln 2x}{h}$

$= \lim_{h \to 0} \frac{\ln(2x+2h) - \ln 2x}{h}$

$= \lim_{h \to 0} \frac{\ln\left(\frac{2(x+h)}{2x}\right)}{h}$

$= \lim_{h \to 0} \frac{\ln\left(\frac{x+h}{x}\right)^{\frac{1}{h}}}{h}$

$\frac{h}{x} = k$

$h = k \cdot x$

$= \lim_{x \to 0} \frac{\ln\left(1 + \frac{h}{x}\right)}{h}$

$= \lim_{k \to 0} \frac{\ln(1+k)}{k \cdot x}$

$$= \lim_{k \to 0} \frac{\ln(1+k)}{k \cdot x}$$

$$= \frac{1}{x} \lim_{k \to 0} \frac{\ln(1+k)}{k}$$

$$\lim_{x \to 0} \frac{\ln(x+1)}{x} = 1$$

$$y'(x) = \frac{1}{x} \Rightarrow \frac{d}{dx} \ln 2x = \frac{1}{x}$$

$$f'(x) = \ln 2x + \cancel{x} \cdot \frac{1}{\cancel{x}} = \ln 2x + 1$$

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} f'(x) = \lim_{x \to 0} \ln 2x + 1 = \lim_{x \to 0} \frac{1}{x} + 0 = \lim_{x \to 0} (-1) \cdot x^{-2} = \infty$$

lim kiri ≠ lim kanan

C. $\lim_{x \to 0} x \ln(2x)$

$u = 2x$

$\frac{u}{2} = x$

$\lim_{u \to 0^+} \frac{u}{2} \ln u = \frac{1}{2} \cdot 0 = 0$ //

$\lim_{x \to 0^+} x \cdot \ln x = 0$

3) Let $f(x) = \begin{cases} \frac{x^2-16}{x-4} & , \text{if } x \neq 4 \\ 8 & , \text{if } x = 4 \end{cases}$

a) Evaluate if the limit exist!

$\lim_{x \to c^+} f(x) = \lim_{x \to c^-} f(x)$ ⟶ $\lim_{x \to 4^+} f(x) = \lim_{x \to 4^-} f(x)$

x	f(x)
5	...
4,9	
4,5	
4,1	
4,01	

x	f(x)
3	...
3,9	...
3,99	...
3,999	...

b) $\lim_{x \to 4} \frac{x^2-16}{x-4} = f(4)$

$\lim_{x \to 4} \frac{(x-4)(x+4)}{(x-4)} = 8$

$4+4 = 8$

$8 = 8$ ✓ kontinu

Syarat 1: $f(4)$ memiliki nilai

$f(4) = 8$

Syarat 2: $\lim_{x \to 4} \frac{x^2-16}{x-4}$ memiliki nilai

$\lim_{x \to 4} f(x) = \lim_{x \to 4} \frac{x^2-16}{x-4} = \lim_{x \to 4} \frac{(x-4)(x+4)}{(x-4)} = 4+4 = 8$

Syarat 3: $f(4) = \lim_{x \to 4} f(x)$

$8 = 8$ ✓ kontinu

• Apakah fungsi $f(x) = \begin{cases} x+1, & untuk\ x < 1 \\ x^2 - x + 2, & untuk\ x \geq 1 \end{cases}$ *apakah Kontinu?*

kiri

kanan

Jawab :

Apakah punya limit ?

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x)$$

$$\lim_{x \to 1^-} (x+1) = \lim_{x \to 1^+} (x^2 - x + 2)$$

$$1+1 = 1^2 - 1 + 2$$

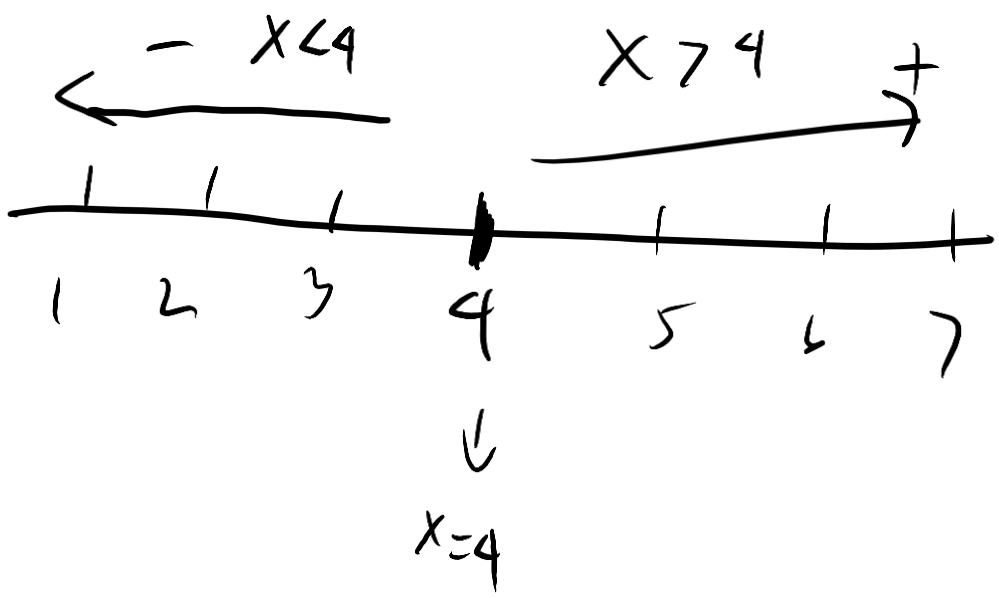
$2 = 2$ memenuhi (punya limit)

$$v = \frac{v_1 + v_2}{1 + \frac{v_1 \cdot v_2}{c^2}}$$

$\lim_{x \to c^-} f(x)$

Mengapa tidak ada limit kanan?
Yapping -> karena dalam Fisika tidak ada benda yg memiliki kecepatan lebih besar dari kecepatan cahaya. Sehingga pendekatan yang bisa dilakukan adalah dari benda yang memiliki kecepatan kurang dari kecepatan cahaya.

Apakah limitnya ada? Untuk saat ini blm ada, karena hanya ada pendekatan dari limit kiri sedangkan limit kanan belum ditemukan.



$x \neq 4 \rightarrow x < 4$ atau $x > 4$

kiri

$$\frac{x^2-16}{x-4}$$

kanan

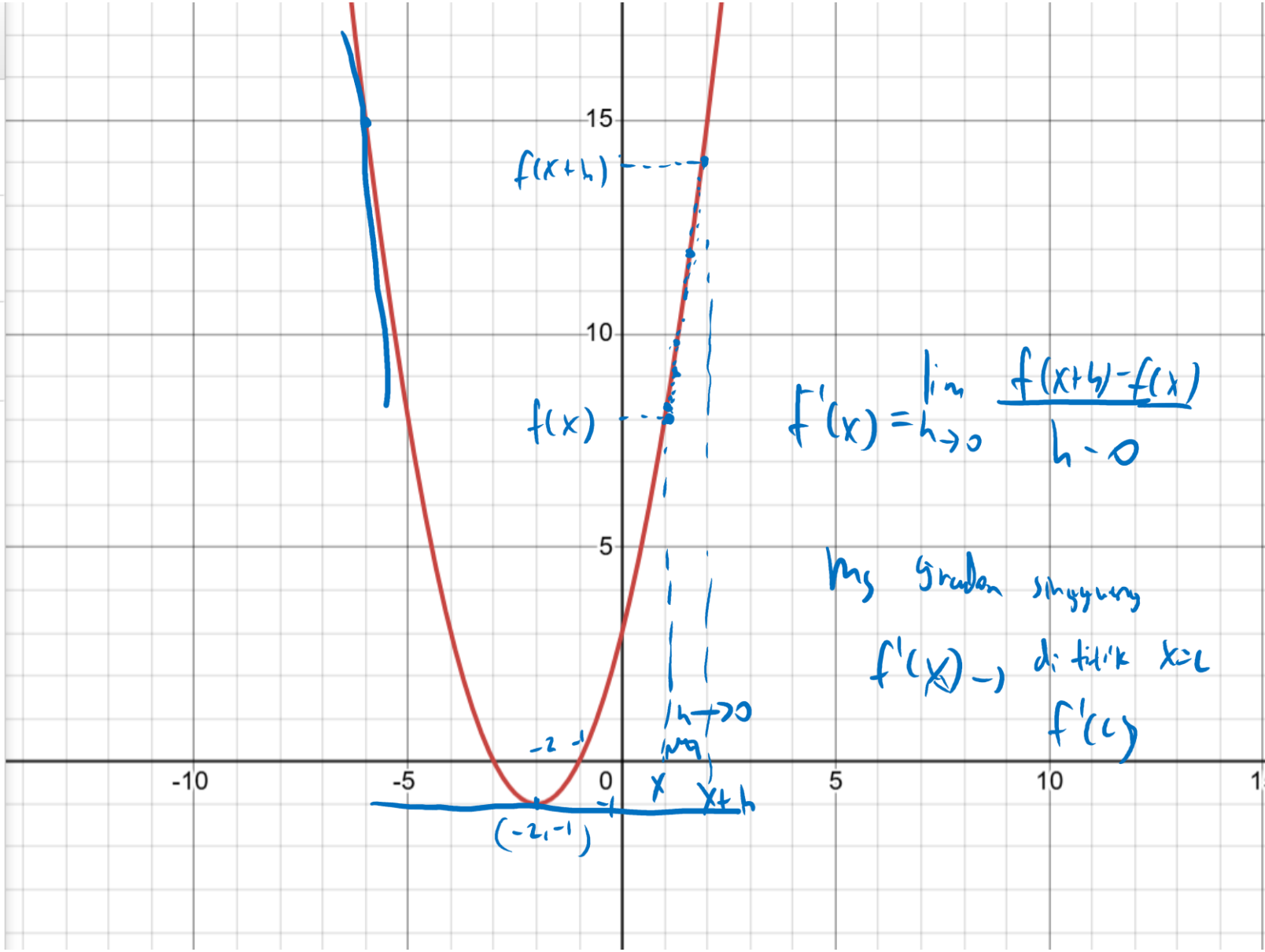
$$\frac{x^2-16}{x-4}$$

1 $f(x) = x^2 + 4x + 3$

2 $g(x) = 2x + 4$

3

4



powered by desmos

Trigon

$\frac{d}{dx} \sin x = \cos x$

$\frac{d}{dx} \cos x = -\sin x$

$\frac{d}{dx} \tan x = \sec^2 x$

$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$

$\frac{d}{dx} \operatorname{cosec} x = -\cot x \cdot \operatorname{cosec} x$

$\frac{d}{dx} \sec x = \tan x \cdot \sec x$

$\frac{d}{dx} u \cdot v = u'v + uv'$

$\frac{d}{dx} \frac{u}{v} = \frac{u'v - uv'}{v^2}$

Invers trigon

$\frac{d}{dx} \arcsin x = ..$

arc

arc

Logaritma

Eksponen

Turunan Implisit

Let $f(x) = 2x^{2} + 4x^{1} + 5$, Evaluate $f'(x)$!

$f'(x) = 2 \cdot 2 \cdot x^{2-1} + 4 \cdot 1 \cdot x^{1-1}$

$f'(x) = 4x + 4$

Let $f(x) = \overbrace{(3x^2 + 2x)}^{u}\overbrace{(x^3 - 1)}^{v}$

$f'(x) = u' \cdot v + u \cdot v'$

$f'(x) = \left(\frac{d}{dx}(3x^2+2x)\right) \cdot (x^3-1) + (3x^2+2x) \cdot \left(\frac{d}{dx}(x^3-1)\right)$

$f'(x) = (6x + 2) \cdot (x^3-1) + (3x^2+2x) \cdot (3x^2)$

$f'(x) = 6x^4 - 6x + 2x^3 - 2 + 9x^4 + 6x^3$

$f'(x) = 15x^4 + 8x^3 - 6x - 2$

Chain's Rule

$$f(x) = \underbrace{\left(5x^4 + 3x^2 + 7x\right)}_{u}^{5}, \quad f'(x) = ?$$

$$y = f(x) = u^5$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{du} = 5 \cdot u^4 = 5 \cdot (5x^4 + 3x^2 + 7x)^4$$

$$\frac{du}{dx} = 5 \cdot 4 \cdot x^3 + 3 \cdot 2 \cdot x + 7 \cdot 1 \cdot x^0 = 20x^3 + 6x + 7$$

$$\frac{dy}{dx} = f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = 5(5x^4 + 3x^2 + 7x)^4 \cdot (20x^3 + 6x + 7)$$

$$f'(x) = 5\left(5x^4 + 3x^2 + 7x\right)^4 \cdot \left(20x^3 + 6x + 7\right)$$

$$f(x) = \tan(3x^2+7x+1) \cdot (\operatorname{cosec}(8x))^3$$

$$f'(x) = \left(\frac{d}{dx}\tan(3x^2+7x+1)\cdot(\operatorname{cosec}(8x))^3 + \tan(3x^2+7x+1)\left(\frac{d}{dx}\operatorname{cosec}^3(8x)\right)\right.$$

$$= \left(\sec^2(3x^2+7x+1)\cdot(6x+7)\right)\cdot \operatorname{cosec}^3(8x) + \tan(3x^2+7x+1)\left(3\cdot\operatorname{cosec}^2 8x\cdot(-\cot 8x\cdot \operatorname{cosec} 8x)\cdot 8\right)$$

$$\frac{d}{dx}(\operatorname{cosec} 8x)^3 = 3(\operatorname{cosec} 8x)^2\cdot(-\cot 8x\cdot\operatorname{cosec} 8x)\cdot 8$$

$y = x + 5$ eksplisit

$x - y + 5 = 0$ ← implisit

Turunan Implisit

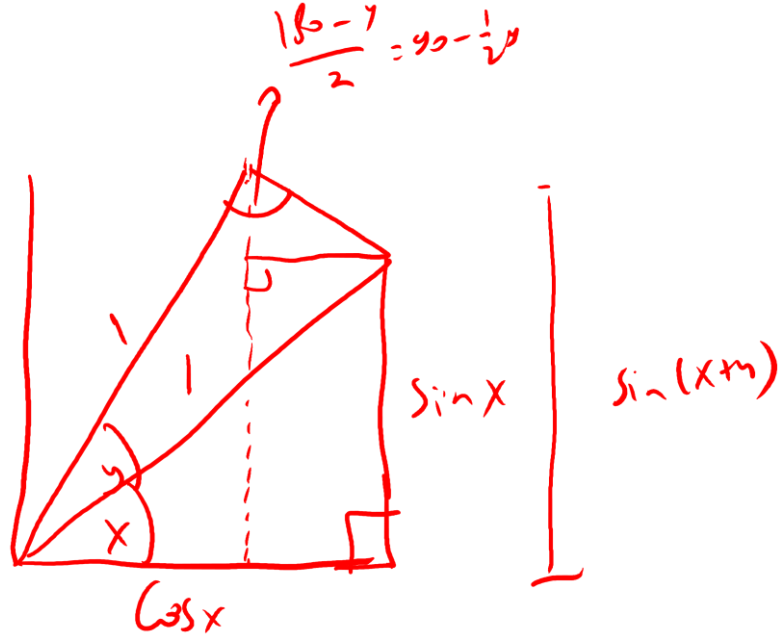
$$If\ x^2 + xy - 2 = 10, find\ \frac{dy}{dx}$$

$2x + x' \cdot y + xy' - 0 = 0$

$2x + y + x \cdot \frac{dy}{dx} = 0$

$(-x)\ \frac{dy}{dx} = 2x + y$

$\frac{dy}{dx} = -\frac{2x+y}{x}$ //


$\frac{180-\gamma}{2} = 90-\frac{1}{2}\gamma$
1
1
γ
x
$\sin x$
$\sin(x+\gamma)$
$\cos x$